

Modular Arithmetic

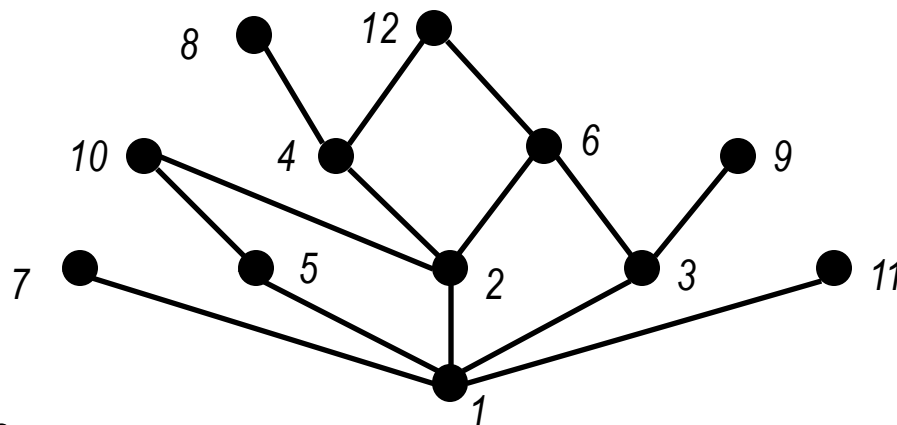
+ finishing primes

Previous Lecture

- *Divisors, properties of divisibility*
- *Greatest common divisor*
- *Euclidean algorithm*

Least Common Multiple

- A positive integer c is called a **common multiple** of integers a and b if $a \mid c$ and $b \mid c$
- The number c is called the **least common multiple** of a and b , denoted $\text{lcm}(a,b)$ if it is a common multiple and for any common multiple d we have $c \mid d$



● Theorem.

For any integers a and b , the least common multiple exists.

Least Common Multiple (cntd)

Theorem

For any integers a and b we have $ab = \text{lcm}(a,b) \cdot \text{gcd}(a,b)$

Example:

Find $\text{lcm}(15,21)$.

The Fundamental Theorem of Arithmetic

● Theorem.

*Every integer $n > 1$ can be represented **uniquely as a product of primes**, up to the order of the primes, called its **prime factorization**.*

● Proof (Existence)

Base case: 2 is prime so it is a product of primes

Inductive step:

Assume every integer $2 \leq i \leq k$ has a prime factorization

Now consider the integer $k+1$

If $k+1$ is prime, we're done

If $k+1$ is not prime, then $k+1=ab$ where $1 < a, b < k+1$

By the inductive hypothesis, a and b have prime factorizations

So $k+1$ also has a prime factorization

Q.E.D

Example

● *Find the prime factorization of 4270*

Uniqueness of prime decomposition

● *To prove uniqueness of the prime factorization, need a lemma*

● *Lemma:*

Let a_i be an integer for $1 \leq i \leq n$, and suppose p is a prime such that $p \mid a_1 a_2 \dots a_n$. Then $p \mid a_i$ for some $1 \leq i \leq n$.

● *Proof of uniqueness:*

Suppose $n = p_1 \dots p_k = q_1 \dots q_m$ are distinct prime factorizations of n

WLOG assume there are no common primes, i.e. $\forall i, j \ p_i \neq q_j$

Then $p_1 \mid p_1 \dots p_k = q_1 \dots q_m$, so by the lemma there exists j such that $p_1 \mid q_j$

But q_j is prime and $q_j \neq p_1$, so we have a contradiction

Relatively Prime

- Numbers a and b such that $\gcd(a,b) = 1$ are called **relatively prime**
- How many relatively prime numbers are there?
- **Euler's totient function** $\varphi(n)$ is the number of numbers k such that $0 < k < n$ and n and k are relatively prime.
- If p is prime then every $k < p$ is relatively prime with p . Hence, $\varphi(p) = p - 1$.
- **Lemma.**
If a and b are relatively prime then $\varphi(ab) = \varphi(a) \cdot \varphi(b)$
- **Corollary.**
If $n = p_1^{s_1} p_2^{s_2} \dots p_u^{s_u}$ is the prime factorization of n , then

$$\varphi(n) = n \cdot \left(1 - \frac{1}{p_1}\right)^{s_1} \left(1 - \frac{1}{p_2}\right)^{s_2} \dots \left(1 - \frac{1}{p_u}\right)^{s_u}$$

Congruences

- *In some situations, we care only about the **remainder** of an integer when it is divided by some number.*
 - *If I bake 79 cookies, how many more do I need to make to sell them in even dozens?*
- ***Congruence modulo m** classifies numbers by their remainder when divided by m*
- *Definition:*

a and b are congruent modulo m if and only if $m \mid (a - b)$
- *We use the notation $a \equiv b \pmod{m}$ to denote a, b congruent modulo m*
- *Fact:*

$a \equiv b \pmod{m}$ if and only if $a = q_1m + r$ and $b = q_2m + r$



same remainder

Congruences (cntd)

● *Examples:*

$$12 \equiv 5 \pmod{7}$$

$$12 \equiv 6 \pmod{3}$$

$$12 \equiv -3 \pmod{15}$$

$$12 \equiv 0 \pmod{12}$$

● *Theorem:*

For any integers a , b , and m , $a \equiv b \pmod{m}$ if and only if $a = b + km$ for some integer k .

Proof:

By definition, $a \equiv b \pmod{m}$ if and only if $a - b = km$ for some integer k . Then $a = b + km$

Congruences and Arithmetic Operations

● *Addition, subtraction, multiplication behave really well with respect to congruences*

● **Theorem.**

Let m be positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

(1) $a + c \equiv b + d \pmod{m}$,

(2) $a - c \equiv b - d \pmod{m}$, and

(3) $ac \equiv bd \pmod{m}$

● *Proof sketch*

(3) Let $a = b + km$ and $c = d + lm$ for some k and l .

*Then $ac = (b + km)(d + lm) = bd + blm + dkm + klm$
 $= bd + (bl + dk + klm)m$*

Example

● $7+11 \pmod{5}$

● $743*987654 \pmod{5}$

Congruences and Arithmetic Operations (cntd)

● *Division is not so good:*

Although $8 \equiv 24 \pmod{4}$ and $4 \equiv 8 \pmod{4}$,

$$\frac{8}{4} = 2 \not\equiv 3 = \frac{24}{8} \pmod{4}$$

Residues

- *Let us consider the binary relation $\equiv$ modulo m on the set of integers, that is,*

$$\{(a,b) \mid a \equiv b \pmod{m}\}$$

- *It is reflexive, symmetric, and transitive. This is an equivalence relation*

- *Each such relation defines a partition on the set of integers into equivalence classes*

$$[a \bmod m] = \{ b \mid a \equiv b \pmod{m} \}$$

- *We choose a **canonical representative** for each such class, which is the (unique) residue of a modulo m*

Residues (cntd)

- The *residue* of an integer a modulo m is such a number b that $a \equiv b \pmod{m}$ and $0 \leq b < m$
- In other words the residue of a modulo m is the remainder of a when divided by m
- Let $\mathbb{Z}_m$ denote the set $\{0, 1, 2, \dots, m-1\}$. This is the set of all possible remainders of integers when divided by m
- It is called the *set of residues*, and its members are called *residues*

Modular Arithmetic

- We define addition, subtraction, and multiplication of residues:

Let $a, b \in \mathbb{Z}_m$. Then

$a + b \pmod{m}$ is the element $c \in \mathbb{Z}_m$ such that $c \equiv a + b \pmod{m}$

$a - b \pmod{m}$ is the element $c \in \mathbb{Z}_m$ such that $c \equiv a - b \pmod{m}$

$a \cdot b \pmod{m}$ is the element $c \in \mathbb{Z}_m$ such that $c \equiv a \cdot b \pmod{m}$

- Example. Construct operation tables for $\mathbb{Z}_5$

+	0	1	2	3	4
0					
1					
2					
3					
4					

.	0	1	2	3	4
0					
1					
2					
3					
4					

Divisors of Zero

- *Modular addition just rotates/shifts/cycles the numbers*
 - *In particular, it is **injective** as a function $f(x) = x + a \pmod{m}$*
- *Multiplication however is **not** injective*

.	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

- A **proper 0 divisor** modulo m is a residue $a \not\equiv 0 \pmod{m}$ such that there exists $b \not\equiv 0 \pmod{m}$ with $a \cdot b \equiv 0 \pmod{m}$.
- $\mathbb{Z}_4$ has a proper 0 divisor. $\mathbb{Z}_5$ does not.

Inverse

● A residue b modulo m is called an **inverse** of a residue a if $a \cdot b \equiv 1 \pmod{m}$, denoted a^{-1}

● 3 is the inverse of 2 modulo 5

● 2 does not have an inverse modulo 4

● Theorem

Let a be residue modulo m . The following conditions are equivalent:

- (i) a has an inverse;
- (ii) a is not a proper divisor of 0;
- (iii) a is **relatively prime** with m .

Inverse (cntd)

● Proof.

(i) $\Rightarrow$ (ii) *By contradiction.*

Suppose $a \cdot b \equiv 0 \pmod{m}$ for some b not equal to 0.

Then $a^{-1} \cdot a \cdot b \equiv a^{-1} \cdot 0 \pmod{m}$

$b \equiv 1 \cdot b \equiv 0 \pmod{m}$, a contradiction

(ii) $\Rightarrow$ (iii) *By contraposition.*

Suppose $\gcd(a, m) = d$ and $a = ld$, $m = kd$.

Then $ak \equiv kld \equiv lm \equiv 0 \pmod{m}$.

Thus a is a proper divisor of 0.

(iii) $\Rightarrow$ (i)

Suppose $\gcd(a, m) = 1$. Then there are u, v with $au + mv = 1$.

Thus $au \equiv 1 \pmod{m}$; a has an inverse.

Practice

Exercises from the Book:

7th edition: 28, 35, 40 (page 245), 5, 17 (page 284 – 285)

8th edition: 34, 41, 46 (page 259), 5, 17 (page 301)